CineBook — Movie Ticket Booking System

Comprehensive Software Architecture & Project Overview

1. Executive Summary

CineBook is an enterprise-grade, responsive Movie Ticket Booking platform designed for cinema operations in Hyderabad, Telangana. Inspired by leading commercial platforms like BookMyShow and District, CineBook implements full-stack software engineering principles, robust ACID transactions, high-concurrency seat locking, and dynamic QR e-tickets.

2. Technology Stack & Frameworks

- **Frontend Framework:** Next.js 16 (App Router) with React 19 and TypeScript 5.9.
- **Design & Styling:** Tailwind CSS v4 with cinema dark navy aesthetic, accessible states, and responsive viewports.
- **Backend & APIs:** Next.js Server Route Handlers with Zod input schema validation and HTTP-only session cookies.
- **Database & ORM:** Prisma ORM 6.4 with SQLite for zero-setup execution, plus full PostgreSQL schema and Docker Compose.
- **Security & Crypto:** jose JWT authentication, bcrypt password hashing, and dynamic qrcode generation.

3. Core Architectural Highlights

- **Concurrency Safe Seat Reservations:** Implements database transactions (prisma.$transaction) to lock selected seats for a 5-minute checkout window. Two concurrent customers attempting to reserve the exact same seat will see one succeed and the second safely rejected with HTTP 409 conflict.
- **Interactive Cinema Seating Grid:** Real-time hall visualizer featuring Regular, Premium, VIP Recliner, and Accessible seating tiers, curvature screen direction, and dynamic polling.
- **Dual Payment Simulation Modes:** Fully functional simulated checkout supporting both Instant Payment Success (ticket confirmation) and Declined Transaction (safe seat rollback).
- **Cryptographic E-Tickets & Gate Pass:** Generates unique cryptographically signed QR codes and verification passes, complete with a cinema usher staff validation portal.

4. Database Entity Relational Model

The CineBook relational schema is organized into 11 distinct models with strict foreign key constraints and cascade rules:

- **User:** Customer and Administrator accounts with encrypted password hashes and profile details.
- **Movie:** Catalog titles with duration, genre, language, certification, cast/crew metadata, and poster URLs.
- **Theatre & Screen:** Multiplex properties across Hyderabad with screen layout configs and amenities.
- **Seat:** Physical seat coordinates (row label, seat number, and category tier).
- **Show:** Movie schedule entries with tiered ticket prices, convenience fees, and 18% GST tax rates.
- **SeatReservation:** Temporary 5-minute atomic seat locks with unique constraint [showId, seatId, status].
- **Booking & BookingSeat:** Confirmed customer ticket orders linked to show, seats, and verification tokens.
- **Payment:** Transaction logs capturing payment status, simulated provider refs, and timestamps.
- **Cancellation:** Tracks cancellations and computes automatic 75% refund policy records.

5. Hyderabad Seed Catalog

The application includes realistic cinema catalog data for immediate demonstration:

- **8 Movies:** Kalki 2898 AD, Pushpa 2: The Rule, Devara: Part 1, Game Changer, HIT: 3rd Case, Sikandar, Spirit, Ramayana.
- **4 Multiplex Theatres:** PVR ICON (Kukatpally), AMB Cinemas (Gachibowli), INOX (Banjara Hills), Prasads Multiplex (Necklace Rd).
- **8 Auditorium Screens:** Dolby Atmos, Laser Atmos, 4DX, and Insignia VIP Recliner halls.
- **872 Seats & 27 Shows:** Fully generated physical seating coordinates and scheduled shows.

6. Automated Verification & Test Results

The test suite executes 10 comprehensive tests covering security, concurrency, transactions, and analytics:

[PASS] Test 1: Password hashing and secure verification with bcrypt
[PASS] Test 2: JWT Session signing and validation with jose
[PASS] Test 3: Duplicate email registration rejection in database
[PASS] Test 4: Role-based access checks (Customer blocked from Admin)
[PASS] Test 5: Successful temporary 5-minute seat reservation
[PASS] Test 6: CONCURRENCY: Second user blocked from reserving same seat (HTTP 409)
[PASS] Test 7: Same physical seat can be booked for different shows
[PASS] Test 8: Expired reservations release seats back to the public pool
[PASS] Test 9: Order creation, 18% GST calculation, and 75% refund processing
[PASS] Test 10: Operational analytics calculation (movies, theatres, shows, revenue)

7. Access Credentials & Quick Start

Pre-configured demonstration accounts available on the login page:

- **Admin Account:** Email: admin@cinebook.com | Password: Admin@123456
- **Customer 1:** Email: arjun@example.com | Password: Customer@123
- **Customer 2:** Email: priya@example.com | Password: Customer@123

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npm install -> npx prisma db push -> npx prisma db seed -> npm test -> npm run dev
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